「Chapter 8.

「Cauchy Goursat Theorem.

Let f be an analytic in an open disk Δ. Then, for any closed curve γ in Δ,

$$\int_\gamma f = 0$$

Proof along the Alfors. First, we will show that: for $R = \{(x,y) \mid a \le x \le b,\ c \le y \le d\}$,

$$\int_{\partial R} f = 0, \quad \text{if } f \text{ is analytic on } R.$$

Denote $\eta(R) \overset{\text{def}}{=} \int_{\partial R} f(z)\,dz$. And, define bisections of R as

$$R^{(1)} = \{(x,y) \mid a \le x \le \tfrac{a+b}{2},\ \tfrac{c+d}{2} \le y \le d\} \quad \text{and}$$

$$R^{(2)} = (\tfrac{b-a}{2}, 0) + R^{(1)},\quad R^{(3)} = (0, -\tfrac{d-c}{2}) + R^{(1)},\quad R^{(4)} = \left(\tfrac{b-a}{2}, -\tfrac{d-c}{2}\right) + R^{(1)}.$$

Then $\eta(R) = \eta(R^{(1)}) + \eta(R^{(2)}) + \eta(R^{(3)}) + \eta(R^{(4)})$, so by the triangle inequality,

$$|\eta(R)| \le \sum_{i=1}^{4} \eta(R^{(i)}) \le \sum_{i=1}^{4} \max_{j=1,\dots,4} \eta(R^{(j)}) = 4 \cdot \eta(R^{(k)}) \quad \text{for some } k = 1,2,3,4$$

Take $R_1 = R^{(k)}$. Repeat this process inductively, we obtain that

$\{R_n\}_{n=0}^{\infty}$ s.t $R_{n+1} \subseteq R_n$ for all $n \ge 0$, $R_n \ne \phi$, R_n compact, and

$$|\eta(R_n)| \ge 4^{-1} \cdot |\eta(R_{n-1})| \ge 4^{-2} |\eta(R_{n-2})| \ge \cdots \ge 4^{-n} \cdot |\eta(R_0)|.$$

Since $\bigcap_{n=0}^{\infty} R_n$ contains exactly one point, denote this z_0, and since f is analytic, given $\varepsilon > 0$, there exists $\delta > 0$ such that $|z - z_0| < \delta \Rightarrow \left| \frac{f(z) - f(z_0)}{z - z_0} - f'(z_0) \right| < \varepsilon$.

Since for all $n \ge 0$, R_n contains z_0, for some $n \in \mathbb{N}$, $R_n \subseteq B_\delta(z_0)$. So,

$$\int_{R_n} f(z)\,dz = \int_{R_n} \big[f(z) \underbrace{- f(z_0) - f'(z_0) \cdot [z - z_0]}_{\int = 0.}\big]\,dz \quad \text{follows that}$$

$$|\eta(R_n)| = \left| \int_{R_n} f(z)\,dz \right| \le \varepsilon \cdot \int_{R_n} |z - z_0| |dz| \le \varepsilon \cdot \operatorname{diam} R_n \cdot \frac{L}{2^n} \quad \text{where } L = 2 \cdot (|b-a| + |d-c|).$$

$$= \varepsilon \cdot \operatorname{diam} R \cdot L \cdot \frac{1}{4^n}.$$

so, $|\eta(R)| \le 4^n \cdot |\eta(R_n)| = \varepsilon L \cdot \operatorname{diam} R \to 0$ as $\varepsilon \to 0$.

Thm. 9.6. [Riemann's Theorem.

Suppose that f has a isolated singularity at $z = z_0$ and it is bounded in $B_R(z_0)\setminus\{z_0\}$ for some $R>0$.

Then there exists a unique analytic extension of f at z_0.

Proof By the assumption, let f be analytic in $0<|z-z_0|\leq R$.

Given z_1 lies in the punctured disk, let $r>0$ so that $r<|z_1-z_0|<R$.

Then, by the Cauchy's Thm, for $C: |z-z_0|=R$ and $C_r: |z-z_0|=r$,

$$f(z_1)=\frac{1}{2\pi i}\cdot\left[\int_C \frac{f(\mu)}{\mu-z_1}\,d\mu-\int_{C_r}\frac{f(\mu)}{\mu-z_1}\,d\mu\right]$$

Meanwhile, since f is bounded, set $M=\sup|f|$, so that

$$\left|\frac{1}{2\pi i}\int_{C_1}\frac{f(\mu)}{\mu-z_1}\,d\mu\right|\leq\frac{M}{2\pi}\cdot\int_{C_1}\frac{1}{|\mu-z_1|}\,|d\mu|.$$

$$\leq\frac{M}{\cancel{2\pi}}\cdot\frac{\cancel{2\pi}r}{|z_1-z_0|-r}\to 0 \quad \text{as } r\to 0.$$

So, $$f(z_1)=\frac{1}{2\pi i}\int_C\frac{f(\mu)}{\mu-z_1}\,d\mu$$

Since the z_1 was arbitrary,

$$f(z)=\frac{1}{2\pi i}\int_C\frac{f(\mu)}{\mu-z}\,d\mu,$$ which is analytic in $|z-z_0|<R$.